Exercise-1

H2C, IOT Project

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The embedded system

An embedded system can be broadly defined as a device that contains tightly coupled hardware and software components to perform a single function, forms part of a larger system, is not intended to be independently programmable by the user and is expected to work with minimal or no human interaction. Two additional characteristics are very common in embedded systems: reactive operation and heavily constrained. Most embedded system interact directly with processes or the environment, making decisions on the fly, based on their inputs. This makes necessary that the system must be reactive, responding in real-time to process inputs to ensure proper operation. Besides, these systems operate in constrained environments where memory, computing power, and power supply are limited. Moreover, production requirements, in most cases due to volume, place high-cost constraints on designs. This is a broad definition that highlights the large range of systems that fall into it. It is controlled by gate arrays, microcontrollers or digital signal processors (DSP), field-programmable gate arrays (FPGA), application-specific integrated circuits (ASIC) and GPU technology. Processing systems are integrated with different modules so that it can handle electric and mechanical interfacing. Embedded systems programming instructions, indicated to as firmware, are stored in flash memory chips. It runs with limited computer hardware resources. Embedded systems connect with the outside world through peripherals, linking input and output devices. (Manuel Jiménez, Rogelio Palomera, Isidoro Couvertier. 2013, 1)

IOT

The Internet of Things (IoT) refers to the network of physical object ‘things’ that are embedded with devices, systems, sensors, gadgets, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet. These devices range from ordinary household objects to sophisticated industrial tools. The main components of the IoT are communication, control and cost savings. This is because IoT devices can collect data and communicate with each other by sharing their data. It can be controlled from any where and this is also cost-effective for home and industry. They process data in the cloud, such as, AWS, Azure and google cloud. They easy to connect over wi-fi and Bluetooth. Some Iot devices, for example, Apple watch, Fitbit, iRobot and Google Home. (Maggie Tillman. 2021)

Industrial iot

The IIoT stands for Industrial Internet of Things, which is the collection of devices, sensors and autonomous instruments connected through the internet to industrial applications. It helps to gather data, carry out analyses and optimise production, increasing the efficiency and it reduces the costs of the manufacturing process and the provision of services. Industrial applications are completely technological eco-friendly that connect devices and these with the people who manage the processes in assembly lines, logistics and large-scale distribution. Currently IIoT applications are primarily focused on energy, manufacturing and transport. It is expected that using of IIoT will increase in the implementation of industrial robots, for example cobots, transport control systems, warehouses and different kinds of maintenance systems. The key differences between IoT and IIoT is that IoT refers to services for consumers, IIoT much more focuses on improving the performance of production sites so that piece of machinery or an entire business process can increase efficiency and safety. (Stephen J. Bigelow. 2021)

Big Data

Big Data refers to a complex structured and unstructured collection of data that is massive in volume and it increases exponentially with time and that is generated from different kinds of sources. It is also data but with massive size. There are 3 types of big data, such as structured, unstructured and semi-structured**.** structured data is called when Any data can be accessed, stored and processed in the fixed format**.** When data is unknown form or the structure is classified is called unstructured data. Semi-structured data can be both forms of data. Three characteristics of Big Data are volume, variety, and velocity. (Siddharth Sonkar. 2020)

Edge AI

Edge AI means that Artificial Intelligence algorithms are processed on a hardware device locally. The algorithms are using sensor data or signals data that are created on the device. Edge AI functional devices does not need to be connected in the network to work properly, it can run itself by processing data and it is able to take decisions independently without a network connection. To use Edge AI, need to use microprocessor and sensors. Edge AI help to do real time operations including decision, action and data creation in a millisecond. Real time operations is very important for AI, such as self-driving cars, robots and many other areas. It will reduce costs for data exchanging or communication because very less data will be transmitted. It will be easy to avoid the problem with streaming and storing a lot of data to the

cloud by processing data locally. Nowadays Edge AI is implemented some real-life applications, for example autonomous car, surveillance and monitoring and industrial Internet of Things(IIoT) .(Softtek. 2020.)

“Big Data” benefit

Big data has remarkably changed the approach of managing and analyzing data across industries. Especially it is noticeable that big data is making big changes is healthcare. Moreover, healthcare analytics has the ability to reduce costs of treatment, prediction of epidemics, avoiding preventable diseases, and improvement the quality of life. The average human life expectancy is increasing because of the revolution of big data analytics. Besides, health professionals are analyzing those data to improve their capability and strategies. Unlike traditional technologies for collecting data, big data in healthcare means that collecting or recording of massive amount of patient data will be saved by the help of digital technologies which will help to increase managing hospital performance. It has so many life-saving outcome. Nowadays people live longer and treatment system has been changed tremendously because of data. Doctors want to know patient’s problem as early as possible to provide more accurate treatment. This is because patient’s data analytics helps more to provide good treatment.

Electronic health/patient records (EHR or EPR) is the most important application of big data in medicine. It is very hard to always modify prescription or remembering all patients’ problem. That is why EHR makes it easier for the doctors. It triggers warnings and reminders to the patients to take service in time. (Vinod Saratchandran. 2019)

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